Chuanwu Zhao, Ph.D.

Postdoctoral Researcher, Beihang University

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EDUCATION

2021.9-2025.6 | Ph.D. in Cartography and Geographic Information Systems, Beijing Normal University, China (Supervisor: Prof. Yaozhong Pan)

2018.9-2021.6 | M.Sc. in Geodesy and Surveying Engineering, China University of Mining and Technology (Beijing), China (Supervisor: Prof. Wei Guo)

2014.9-2018.6 | B.Sc. in Surveying and Mapping Engineering, Henan Polytechnic University, China (Thesis Supervisor: Prof. Changchun Li)

EMPLOYMENT

2025.7-2027.6 (expected) | Postdoc in Remote Sensing Ecology, Beihang University, China (Supervisor: Prof. Huijie Zhao and Prof. Guifei Jing)

RESEARCH INTERESTS

Forest disturbance ecology, vegetation dynamics, conservation effectiveness, forest carbon dynamics, global change ecology, and ecological remote sensing.

PUBLICATIONS AND ONGOING PROJECTS

Peer-reviewed publications (* corresponding authors)

First and corresponding author publications

Published:

1. **Zhao C***, Zhao H, Pan Y, Jing G. Optimizing Spectral Inputs for Deep Learning-Based Vegetation Destruction Detection Using Sentinel-2 Imagery. *Remote Sensing*. 2026, 18(14), 2333.
2. **Zhao C**, Pan Y*, Ren S, Gao Y, Wu H, Ma G. Accurate vegetation destruction detection using remote sensing imagery based on the three-band difference vegetation index (TBDVI) and dual-temporal detection method. I*nternational Journal of Applied Earth Observation and Geoinformation*, 2024, 127, 103669.
3. **Zhao C**, Pan Y*, Wu H, Ren S, Ma G, Gao Y, Zhu Y, Jing G. A novel spectral index for vegetation destruction event detection based on multispectral remote sensing imagery. *IEEE Journal of Selected Topics in Applied Earth Observations and Remote Sensing*, 2024, 17, 11290-11309.
4. **Zhao C**, Pan Y*, Zhang P. Development of a new indicator for identifying vegetation destruction events using remote sensing data. *Ecological Indicators*, 2024, 166, 112553.
5. **Zhao C**, Pan Y*, Zhu X, Li L, Xia X, Ren S, Gao Y. Monitoring of deforestation events in the tropics using multidimensional features of Sentinel 1 radar data. *Frontiers in Forests and Global Change*, 2023, 6, 1257806.

6. **Zhao C**, Pan Y*, Wu H, Zhu Y. A new effective and robust index using spectral curve shapes and local reflectance peaks: detecting simple coated steel-tiled houses in urban areas. *IEEE Transactions on Geoscience and Remote Sensing*, 2025, 63, 4404716.
7. **Zhao C**, Wei H, Feyisa GL, de Castro Tayer T, Ma G, Wu H, Pan Y*. Evaluating spectral indices for water extraction: limitations and contextual usage recommendations. *International Journal of Applied Earth Observation and Geoinformation*, 2025, 139, 104510.
8. **Zhao C**, Pan Y*, Wu H, Zhu Y. Quantifying the contribution of industrial zones to urban heat islands: Relevance and direct impact. *Environmental Research*, 2024, 240, 117594.
9. **Zhao C**, Pan Y*. A novel spectral index for mapping blue colour-coated steel roofs (BCCSRs) in urban areas using Sentinel-2 data. *International Journal of Digital Earth*, 2023, 16(1), 2862-2884.
10. **Zhao C**, Pan Y*, Teng Y, Baqa MF, Guo W. Air quality improvement in China: evidence from PM2.5 concentrations in five urban agglomerations, 2000–2021. *Atmosphere*, 2022, 13(11), 1839.
11. **Zhao C**, Zhang Y*, Guo W, Baqa MF. Dynamics and drivers of water clarity derived from Landsat and in-situ measurement data in Hulun lake from 2010 to 2020. *Water*, 2022, 14(8), 1189.
12. Guo W, **Zhao C***, Zhang Y, Gao S. Mapping impervious surface distribution and dynamics in an arid/semiarid area-A case study in Ordos, China. *IEEE Access*, 2021, 9, 19659-19673.
13. **Zhao C**, Wu Z, Zhang Y, Guo W*. Analysis of impervious surface change in Ordos city based on multi-temporal remote sensing. Bulletin of Surveying and Mapping, 2022, (5), 56–61.
14. **Zhao C**, Guo W*, Yan Y, Dai H, Zhang J. Analysis of Vegetation Cover Change and Driving Forces in Typical Resource. Remote Sensing Technology and Application, 2022, 37(2), 389–398.

In Review:

15. **Zhao C***, Zhao H, Pan Y, Yang D, Jing G, Zhao L, Wu H, Zhao F, Seidl R. Climate-related forest disturbances increasingly shape protected areas worldwide. ***Nature Communications***. In review.

<u>*Co-author publications*</u>

Published:

1. Ma G, **Zhao C**, Wu H, Gao Y, Ren S, Pan Y*. A novel red-edge-based anomalous water detection index and diagnostic framework. *International Journal of Digital Earth*, 2025, 18(2), 2540078.
2. Wu H, **Zhao C**, Zhu Y, Pan Y*. A multiscale examination of heat health risk inequality and its drivers in mega-urban agglomeration: A case study in the Yangtze River Delta, China. *Journal of Cleaner Production*, 2024, 458, 142528.
3. Ren S, Pan Y*, **Zhao C**, Gao Y, Ma G. ASAI: A general and training-free artificial surfaces anomaly index using post-disaster single-temporal and high-resolution imagery. *International Journal of Applied Earth Observation and Geoinformation*, 2026, 146, 105050.
4. Tang M, Zhu X, **Zhao C**, Song J, Xiao C, Guo R. REBAI: Development and validation of a novel indicator for burned area detection using Sentinel-2 images. *Ecological Indicators*, 2026, 182, 114501.
5. Ji X, Sun Y, Guo W*, **Zhao C**, Li K. Land use and habitat quality change in the Yellow River Basin: A perspective with different CMIP6-based scenarios and multiple scales. *Journal of Environmental Management*, 2023, 345(12), 118729.
6. Cui Y, Shi G, **Zhao C**, Li Q, Yao H, Guo W*. A global continuous 500 m nighttime light dataset (1992-2024) via NDVI-guided DMSP-OLS correction and U-TransNet cross-sensor harmonization. *International Journal of Applied Earth Observation and Geoinformation*, 2026, 148, 105230.

7. Guo W, Teng Y*, Yan Y, **Zhao C**, Zhang W, Ji X. Simulation of land use and carbon storage evolution in multi-scenario: A case study in Beijing-Tianjin-Hebei urban agglomeration, China. *Sustainability*, 2022, 14(20), 13436.
8. Ren S, Pan Y*, Zhu X, **Zhao C**, Gao Y. A general and simple automated impervious surface mapping approach based on three-dimensional texture features (3DTF) using fine spatial resolution remotely sensed imagery. *Science of The Total Environment*, 2024, 923, 171181.
9. Zhu Y, Pan Y*, Zhang D, Wu H, **Zhao C**. A Deep Learning Method for Cultivated Land Parcels' (CLPs) Delineation From High-Resolution Remote Sensing Images With High-Generalization Capability. *IEEE Transactions on Geoscience and Remote Sensing*, 2024, 62, 441525.
10. Zhu Y, Pan Y*, Hu T, Zhang D, **Zhao C**, Gao Y. A generalized framework for agricultural field delineation from high-resolution satellite imageries. *International Journal of Digital Earth*, 2024, 17(1), 2297947.
11. Guo W, Teng Y*, Li J, Yan Y, **Zhao C**, Li Y, Li X. A new assessment framework to forecast land use and carbon storage under different SSP-RCP scenarios in China. Science of The Total Environment, 2024, 912, 169088.
12. Gao Y, Pan Y*, Zhu X, Li L, Ren S, **Zhao C**, Zheng X. FARM: A fully automated rice mapping framework combining Sentinel-1 SAR and Sentinel-2 multi-temporal imagery. *Computers and Electronics in Agriculture*, 2023, 21, 108262.
13. Gao Y, Pan Y*, Zhu X, Wu H, Ren S, **Zhao C**. NDCI-mGMM: a novel and automated model for dynamic maize mapping during the growing season. *GIScience & Remote Sensing*, 2026, 63(1), 2606468.
14. Wang L, Gao Y, Pan Y*, Feng W, Xia X, **Zhao C**, Zheng X, Che H, Huang Y, Hou X. A novel spectral and phenological composite index for the early automatic mapping of potatoes from Sentinel-2 multi-temporal images. *Computers and Electronics in Agriculture*, 2026, 244, 111459.

Seminars:

1. **Zhao C**, Pan Y*, Ren S, Ma G, Gao Y, Wu H, Zhu Y. Improving the detection accuracy of vegetation destruction events using bands sensitive to vegetation foliage, canopy and water content. *EGU General Assembly 2024*, Vienna, Austria, EGU24-1217.
2. Ren S, Pan Y*, **Zhao C**, Gao Y, Ma G, Wu H, Zhu Y, Zhang Z. A novel artificial surface anomaly index (ASAI) based on post-disaster texture features using single-temporal and high-resolution imagery. *EGU General Assembly 2024*, Vienna, Austria, EGU24-1228.
3. Wu H, Pan Y*, Gao Y, Jin D, **Zhao C**, Ren S. Improving sample applicability for early-season mapping of winter wheat using geoclimatic zoning. *IGARSS 2024 - 2024 IEEE International Geoscience and Remote Sensing Symposium*, Athens, Greece.
4. Gao Y, Pan Y*, Ren S, **Zhao C**. Comparison of multi-texture features extracts from GF-1 images to map agricultural plastic greenhouse through an object-based approach. *IGARSS 2023 - 2023 IEEE International Geoscience and Remote Sensing Symposium*, California, United States.

Research projects:

1. Joint Optical and Radar Remote Sensing for Detecting Abrupt Vegetation Destruction Events: Methods and Case Studies. China Postdoctoral Science Foundation, 78th General Funding Round. January 2026–December 2027. (PI).
2. Deforestation Detection Using Sentinel-1 Data. 2022 Open Research Fund, State Key Laboratory of Remote Sensing and Digital Earth and Beijing Engineering Research Center for Land Surface Remote Sensing Data Products. July 2022–January 2024. (PI).

3. Remote Sensing Response Characteristics and Semantic Representation of Land Surface Anomalies. Subproject of a Major Project funded by the National Natural Science Foundation of China (NSFC). January 2022–December 2026. (PI: Prof. Yaozhong Pan.)

ACADEMIC SERVICE

Guest editor:

The special issue "Monitoring Vegetation Dynamics Using Remote Sensing Images", Remote Sensing

Reviewer for journal:

PLOS ONE, Ecology and Evolution, Geo-spatial Information Science, International Journal of Digital Earth, GIScience & Remote Sensing, Remote Sensing, Land, Sustainability, IEEE Transactions on Geoscience and Remote Sensing, IEEE Journal of Selected Topics in Applied Earth Observations and Remote Sensing

PROFESSIONAL SOCIETIES

Professional membership/participant: IEEE Geoscience and Remote Sensing Society, Zhejiang Province Geographical Society, Zhejiang Province Remote Sensing Society

SCHOLARSHIPS AND AWARDS

Selected Scholarship:

Graduate Student Academic Innovation, Beijing Normal University, CNY10,000, 2025

Graduate Student Academic Innovation, Beijing Normal University, CNY20,000, 2024

Academic Scholarship, Beijing Normal University, CNY18,000, 2022, 2023

Academic Scholarship, China University of Mining and Technology (Beijing), CNY8,000, 2020

Academic Scholarship, China University of Mining and Technology (Beijing), CNY10,000, 2019

Academic Scholarship, Henan Polytechnic University, CNY1,500, 2015

Selected Awards:

Beijing Outstanding Graduate Awards, Beijing Municipal Commission of Education, China, 2025

Outstanding Graduate Awards, Beijing Normal University, 2025

Merit Student, Beijing Normal University, 2023

Beijing Outstanding Graduate Awards, Beijing Municipal Commission of Education, China, 2021

Outstanding Graduate Awards, China University of Mining and Technology (Beijing), 2021

School Outstanding Student, China University of Mining and Technology (Beijing), 2019, 2020

Merit Student, Henan Polytechnic University, 2015

PROFESSIONAL SKILLS

Programming: Python, R, MATLAB

Remote Sensing and GIS: Optical and SAR Data Processing, Satellite Image Time-Series Analysis, Change Detection, Geospatial Data Analysis

Computational Methods: Statistical Modeling, Machine Learning, Deep Learning, High-Performance Computing

Software and Platforms: Google Earth Engine, ArcGIS, ENVI, SPSS